VOLUME 2B — THEORY NOTES: LOGICAL REASONING, QUANTITATIVE APTITUDE, VERBAL ABILITY

> **Weight check:** these three areas are **LOW–MEDIUM priority** for DAT 2027. They appeared as a handful of MCQs in older papers and were absent in the reported 2026 paper. Keep this to about 10–15% of study time. They still help indirectly: logical structure helps problem-solving answers, and good writing helps the 200-word tasks.

PART 1: LOGICAL REASONING

1.1 Series (number and letter)

Concept: find the rule that generates each term.
Common rules:

- constant difference (+3)
- increasing difference (+1, +2, +3...)
- multiplication (×2)
- squares or cubes (1, 4, 9, 16; 1, 8, 27)
- primes (2, 3, 5, 7, 11)
- alternating two series
- Fibonacci (each term = sum of the previous two)

Letter positions: A1 B2 C3 D4 E5 F6 G7 H8 I9 J10 K11 L12 M13 N14 O15 P16 Q17 R18 S19 T20 U21 V22 W23 X24 Y25 Z26.
Memory aid **"EJOTY"** = 5, 10, 15, 20, 25. Opposite letters sum to 27 (A↔Z, B↔Y, M↔N).

Trap: stopping at the first rule that fits only 2–3 terms. Check that it holds for all terms.
How NID tests it: occasionally, often disguised visually (dots increasing in a series of figures).

Practice

1. 2, 6, 12, 20, 30, ? (a) 40 (b) 42 (c) 44 (d) 36
2. 1, 1, 2, 3, 5, 8, ? (a) 11 (b) 12 (c) 13 (d) 15
3. B, E, H, K, ? (a) M (b) N (c) O (d) L
4. 3, 9, 27, 81, ? (a) 162 (b) 243 (c) 324 (d) 108

Answers: 1-b, 2-c, 3-b, 4-b.
Explanations: (1) The differences 4, 6, 8, 10 give 12 next, so 30 + 12 = 42 (equivalently n(n + 1)). (2) Fibonacci: 5 + 8 = 13. (3) +3 each time: K(11) + 3 = N(14). (4) ×3: 81 × 3 = 243.
Difficulty: Easy · **Priority:** LOW

1.2 Analogies & Classification (Odd One Out)

Analogy: A : B :: C : ?. Find the relationship between A and B (part–whole, tool–user, product–material, cause–effect, worker–tool, degree), then apply it.

Design-flavoured analogies are typical for NID:

- Potter : Wheel :: Weaver : **Loom**
- Pashmina : Kashmir :: Muga : **Assam**
- Bauhaus : Gropius :: De Stijl : **Mondrian / van Doesburg**
- Pixel : Image :: Brick : **Wall** (unit : whole)

Classification: find the item that breaks the shared property. Check several properties: category, material, region, era, function.

Trap: finding a surface similarity and missing the deeper one, e.g. treating the order of terms as irrelevant (A:B relationship reversed).

Practice

1. Kalamkari : Andhra Pradesh :: Phulkari : ? (a) Gujarat (b) Punjab (c) Bihar (d) Odisha
2. Odd one out: Madhubani, Warli, Gond, Kanjeevaram (a) Madhubani (b) Warli (c) Gond (d) Kanjeevaram
3. Chisel : Sculptor :: Scalpel : ? (a) Butcher (b) Surgeon (c) Carpenter (d) Barber
4. Odd one out: Helvetica, Futura, Garamond, Arial

Answers: 1-b, 2-d, 3-b, 4 → Garamond.
Explanations: (1) Phulkari is a Punjabi embroidery. (2) Kanjeevaram is a silk sari (textile); the others are painting traditions. (3) The tool–user relation. (4) Garamond is a serif; the others are sans serif.
Difficulty: Easy–Medium · **Priority:** MEDIUM (design analogies)

1.3 Coding–Decoding

Types: letter shift (+1, −2), reverse alphabet (A↔Z), number substitution, symbol substitution.
Method: compare one word with its code letter by letter and write the shift.

Practice: If DESIGN is coded as EFTJHO, how is CRAFT coded?
Answer: each letter +1, so C→D, R→S, A→B, F→G, T→U = **DSBGU**.
Difficulty: Easy · **Priority:** LOW

1.4 Syllogisms (Deduction)

Rules:

- "All A are B" + "All B are C" → "All A are C" ✓.
- "Some A are B" does **not** give "Some B are not A".
- From two "some" statements, **no** definite conclusion follows.
- From "All A are B", "Some B are A" follows ✓.

Method: draw Venn diagrams. A conclusion must be true in **every** possible diagram.

Practice: Statements: All designers are observers. Some observers are painters. Conclusions: I. Some designers are painters. II. Some observers are designers.
(a) Only I (b) Only II (c) Both (d) Neither
Answer: (b). II follows: all designers are observers, so some observers are designers. I is only possible, not certain.
Difficulty: Medium · **Priority:** LOW

1.5 Statements, Assumptions, Inferences, Cause–Effect

- **Assumption:** an unstated idea the statement takes for granted.
- **Inference:** a conclusion that probably follows from the facts.
- **Cause–effect:** decide which event led to the other. A shared third cause is possible.

Practice: Statement: "Use biodegradable cups at the fest to reduce waste." Assumption: I. Biodegradable cups produce less lasting waste. II. All students will refuse plastic cups.
Answer: Only I is assumed. II is extreme and not needed.
Difficulty: Easy–Medium · **Priority:** LOW

1.6 Arrangements & Puzzles

Method:

1. Draw slots.
2. Place the fixed clues first.
3. Use "not" clues to eliminate.
4. Try cases for the remaining clues.

Practice: Five artists A, B, C, D, E sit in a row facing north. C is in the middle. A is at the left end. B is to the right of C, but not at the right end. E is not next to A. Who is at the right end?
Solution: positions 1–5. A = 1, C = 3. B is right of C and not at 5, so B = 4. The remaining spots are 2 and 5 for D and E. E is not next to A, so E ≠ 2. So E = 5 and D = 2.
Answer: E.
Difficulty: Medium · **Priority:** LOW

1.7 Directions

North is up. Turning right from North = East. Use Pythagoras for the shortest distance.

Practice: A student walks 3 km North, then 4 km East. How far is the student from the start?
Answer: √(9 + 16) = **5 km** (North-East).
Difficulty: Easy · **Priority:** LOW

PART 2: QUANTITATIVE APTITUDE (NID-relevant only)

How much maths does NID really need? Class 8–10 arithmetic and mensuration, used in **design contexts**: scaling a drawing, tiles on a floor, fabric needed, ratios on a map, percentages in a survey, packaging volume. A strong student needs about 6–8 hours of maths revision in total. Stop there.

2.1 Percentages

- x% of y = xy/100. Percentage change = (change ÷ original) × 100.
- Successive changes a% and b%: net = a + b + ab/100. For +20% then −20%: net = −4%.

Practice: A ₹800 lamp is discounted 15%. Find the price. → 800 × 0.85 = **₹680**.

2.2 Ratio & Proportion; Scale

- If a : b = 2 : 3 and the total is 50, the parts are 20 and 30.
- **Map/drawing scale:** 1 : 100 means 1 cm on paper = 100 cm (1 m) in reality.

Practice: On a 1 : 50 drawing, a table is 3 cm long. Find the real length. → 3 × 50 = 150 cm = **1.5 m**.

2.3 Averages

Average = sum ÷ count. If one value changes by d, the average changes by d/n.

Practice: The average of 5 marks is 60. One mark of 40 is replaced by 90. Find the new average. → the change is +50 over 5 = +10, so **70**.

2.4 Fractions & Unit Conversion

- 1 inch = 2.54 cm; 1 foot = 12 in = 30.48 cm; 1 m = 100 cm; 1 km = 1000 m; 1 L = 1000 cm^3; 1 m^2 = 10,000 cm^2; 1 hectare = 10,000 m^2.
- 1 kg = 1000 g. Paper: A4 = 210 × 297 mm. A3 = 297 × 420 mm (twice A4). The aspect ratio of A-series paper is 1 : √2 ≈ 1 : 1.414.

Practice: How many A4 sheets make one A1? → A1 = 2 A2 = 4 A3 = **8 A4**.

2.5 Time, Speed, Distance

- Distance = speed × time. To convert km/h to m/s, multiply by 5/18.
- Clock angle: θ = |30H − 5.5M| degrees.

Practice

1. A train at 72 km/h. Find its speed in m/s. → 72 × 5/18 = **20 m/s**.
2. The angle between the hands at 3:30? → |90 − 165| = **75°**.

2.6 Geometry & Mensuration

Shape	Perimeter / surface	Area / volume
Square (side a)	4a	a^2
Rectangle	2(l + b)	lb
Triangle	sum of sides	½ × base × height
Circle (radius r)	2πr	πr^2
Cube (a)	surface $6a^2$	volume a^3
Cuboid	surface 2(lb + bh + hl)	volume lbh

Shape	Perimeter / surface	Area / volume
Cylinder	curved surface $2\pi rh$; total $2\pi r(r + h)$	volume $\pi r^2 h$
Cone	slant $l = \sqrt{(r^2 + h^2)}$; curved surface πrl	volume $\frac{1}{3}\pi r^2 h$
Sphere	surface $4\pi r^2$	volume $\frac{4}{3}\pi r^3$

- Angles in a triangle sum to 180°; in a quadrilateral to 360°. The sum of interior angles of an n-gon is (n − 2) × 180°. Each interior angle of a regular hexagon = 120°.
- **Tessellation:** among regular polygons, only equilateral triangles, squares and hexagons tile a plane alone (because 60°, 90° and 120° divide 360° exactly).
- Scaling: if lengths are scaled by k, area scales by k^2 and volume by k^3.

Practice

1. A 4 m × 3 m floor is tiled with 50 cm × 50 cm tiles. How many tiles? → (400 × 300) ÷ (50 × 50) = 120000 ÷ 2500 = **48**.
2. A model is built at half scale. How does its volume compare with the original? → $(\frac{1}{2})^3 = \frac{1}{8}$.
3. Why do bee hives use hexagons? → Hexagons tile with no gaps and enclose the most area for a given perimeter among shapes that tile, so they save wax.

2.7 Logical-Mathematical Puzzles

- Handshakes among n people: n(n − 1)/2.
- Number of diagonals of an n-gon: n(n − 3)/2.
- Pigeonhole principle: with n boxes and n + 1 items, at least one box holds 2.

Practice

1. 10 people each shake hands with every other person once. How many handshakes? → 10 × 9 / 2 = **45**.
2. How many diagonals does a hexagon have? → 6 × 3 / 2 = **9**.

Difficulty (Part 2 overall): Easy · **Priority:** LOW

PART 3: VERBAL ABILITY

3.1 Vocabulary for design (learn these; they also help in writing)

Word	Meaning
Aesthetic	concerned with beauty or appearance
Ergonomic	designed for efficiency and comfort in use
Utilitarian	designed to be useful rather than attractive
Ornate	elaborately decorated
Minimalist	using few elements; simple
Juxtapose	place side by side for contrast
Motif	a repeated decorative design or idea
Vernacular	local and traditional (vernacular architecture = local materials and building methods)
Iconography	visual images and symbols used in a work and their meaning

Word	Meaning
Palette	a range of colours used
Tactile	relating to touch
Ephemeral	lasting a very short time
Ubiquitous	found everywhere
Frugal	economical; simple and low-cost
Intuitive	easy to understand without explanation
Prototype	an early model for testing
Iterate	repeat a process to improve
Monochrome	in one colour
Symmetry	identical on both sides of an axis
Kitsch	art or objects considered tasteless due to garish excess, though sometimes appreciated ironically
Avant-garde	new and experimental
Pastiche	an imitation of the style of another work or period
Ambient	relating to the surrounding environment
Patina	a surface sheen caused by age (e.g. green on bronze)
Filigree	delicate metal ornamental work

3.2 Synonyms & Antonyms

Practice

1. Synonym of *ubiquitous*: (a) rare (b) omnipresent (c) unique (d) hidden
2. Antonym of *ornate*: (a) decorative (b) plain (c) elaborate (d) intricate
3. Synonym of *ephemeral*: (a) permanent (b) transient (c) heavy (d) ancient

Answers: 1-b, 2-b, 3-b.

3.3 Idioms (common in older GAT sections)

Idiom	Meaning
Think outside the box	think creatively
Back to the drawing board	start again after a failure
A picture is worth a thousand words	visuals communicate efficiently
Form follows function	design should arise from purpose (Louis Sullivan, 1896)
Less is more	simplicity is powerful (associated with Mies van der Rohe)
Less is a bore	Robert Venturi's postmodern reply
Cut corners	do something poorly to save time or money
The devil is in the details	small things matter
Reinvent the wheel	waste effort recreating something that exists

3.4 Reading Comprehension: method

1. Read the questions first (30 sec).
2. Read the passage for its main idea and tone.

3. Answer from the text, not from your opinion.
4. Eliminate extreme options ("always", "never").

Mini passage: "Traditional Indian stepwells were not merely water sources; they were social spaces where women gathered, travellers rested and festivals were held. Their descending architecture kept the water cool and the air several degrees lower than the surface. Many fell into disuse when piped water arrived, and some became dumping grounds."

Questions & answers

1. The main idea is: (a) stepwells were only for water (b) stepwells served social and climatic functions and declined with piped water (c) piped water is harmful (d) stepwells are dumping grounds. **Ans: (b)**. It covers all three parts of the passage.
2. "Descending architecture" helped mainly by: **keeping the water and air cool**. This is stated directly in the passage.
3. Tone: **informative, mildly regretful**. "Fell into disuse" and "dumping grounds" signal loss.

3.5 Descriptive & Persuasive Writing (HIGH priority: used in the subjective part)

Template for "Draw a product you love and write about it in 200 words" (asked in DAT 2026):

1. **Hook (1–2 lines):** what it is and the emotional connection.
2. **Form & material (2–3 lines):** shape, size, colour, texture, material.
3. **Function & use (2–3 lines):** how it works; the user experience.
4. **Design qualities (2–3 lines):** what is good: ergonomics, durability, simplicity, cultural value.
5. **One improvement (1–2 lines):** this shows critical thinking.
6. **Closing (1 line):** why it matters.

Model answer (about 190 words): "My steel tiffin box"

> My three-tier stainless-steel tiffin has travelled with me to school for nine years, and it still looks almost new. Its form is simple: three stacked cylindrical containers held together by a single clip-on frame with a handle at the top. The brushed steel surface hides scratches, and the rounded edges make it easy to clean.
> What I love most is how honestly it works. Each tier keeps food separate, so dal never leaks into the rotis. The single clamp locks all three tiers at once, and the handle lets me carry the whole lunch with one hand on a crowded bus. Nothing is decorative; every part has a purpose.
> It is also sustainable. It needs no plastic wrap, it never breaks, and it can be repaired or recycled. It echoes the dabbawalas of Mumbai, who move thousands of similar tiffins every day with remarkable accuracy.
> If I could improve one thing, I would add a silicone gasket to each lid so that gravies do not spill when the bus brakes suddenly.
> For me, this tiffin shows that good design can be humble, durable and deeply connected to everyday Indian life.

Writing rules: use concrete nouns, sensory adjectives and active verbs; stay within ±10% of the word limit; write a short heading; keep handwriting legible.

Difficulty: Easy–Medium · **Priority:** HIGH (writing) · LOW (grammar MCQs)

NEXT VOLUME: Volume 2C — General Awareness, Design Awareness, Indian Art & Culture, Architecture (`02C-VOL2-THEORY-GK-ART-ARCHITECTURE.md`)